Curriculum Vitae

JYOTSNA GIDUGU

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INTERESTS

I am a PhD student at University of California San Diego, interested in the area of condensed matter theory. Specifically, I am interested in studying open quantum systems, non-equilibrium many-body physics and quantum information.

RESEARCH SUMMARY

I am currently working on studying the **effects of dissipation in open quantum systems**. I have worked on projects that study the effects of dissipation on spin systems and fermionic systems. I had previously worked on **many-body localization in quantum systems** as a part of my Bachelor's dissertation and on **Josephson junction arrays** as a part of my Master's dissertation. I had also worked on **experimental solid state physics** projects during the first three years of my undergraduate study.

EDUCATION

Currently pursuing a PhD in physics at **University of California San Diego (UCSD)** — 2020-Present

Completed a Master of Science degree at the **Indian Institute of Science (IISc), Bengaluru, India** — 2019 - 2020

Completed a Bachelor of Science (Research) with a major in Physics and a minor in Chemistry at the **Indian Institute of Science (IISc), Bengaluru, India** — 2015 - 2019

TEACHING EXPERIENCE

Served as an instructor at UCSD for:
PHYS 1B (Electricity and Magnetism) during Summer Session 1, 2024. This was done through the Summer Graduate Teaching Scholars (SGTS) program at UCSD. The program included training (Course: Introduction to College Teaching), learning to plan my course and classroom observation during instruction. Mentor: Prof. Philbert Tsai.

Served as a Lab teaching assistant coordinator (LTAC) at UCSD for:
Physics 1-series lab courses for seven quarters and one summer session.

Served as a Teaching Assistant at UCSD for:
PHYS 1BL (Electricity and Magnetism Laboratory- online), 1AL (Mechanics Laboratory- online), 1CL (Waves, Optics, and Modern Physics Laboratory), 2C (Fluids, Waves, Thermodynamics, and Optics), 2CL (Electricity and Magnetism Laboratory), 4D (Electromagnetic Waves, Optics, and Special Relativity), 100B (Electromagnetism II), 110A (Mechanics I), 130A (Quantum Physics I), 140A (Statistical Physics).

PREPRINTS

In preparation- **Jyotsna Gidugu**, Wei-Ting Kuo, Daniel P Arovas. Effects of dissipation and dephasing on the Aubry-André-Harper model.

PUBLICATIONS

Jyotsna Gidugu and Daniel P. Arovas. Dissipative Dirac matrix spin model in two dimensions. Phys. Rev. A 109, 022212 – Published 9 February, 2024.

Soumi Ghosh, **Jyotsna Gidugu**, and Subroto Mukerjee. Transport in the nonergodic extended phase of interacting quasiperiodic systems. Phys. Rev. B 102, 224203– Published 23 December 2020.

Pranava K. Sivakumar, Börge Göbel, Edouard Lesne, Anastasios Markou, **Jyotsna Gidugu**, James M. Taylor, Hakan Deniz, Jagannath Jena, Claudia Felser, Ingrid Mertig, and Stuart S. P. Parkin. Topological Hall Signatures of Two Chiral Spin Textures Hosted in a Single Tetragonal Inverse Heusler Thin Film. ACS Nano 2020, 14, 13463-13469.

Krishna Murthy J, **Jyotsna G**, N Nileena, Anil Kumar PS. Strain induced ferromagnetism and large magnetoresistance of epitaxial $La_{1.5}Sr_{0.5}CoMnO_6$ thin films. Journal of Applied Physics. 2017 Aug 14;122(6):065307.

CONFERENCES

Gave a short talk at the **2025 APS Global Physics Summit, Anaheim**. Effects of dissipation and dephasing on the Aubry-André-Harper model, Jyotsna Gidugu, Wei-Ting Kuo, Daniel P Arovas. — Mar 16-21, 2025

Attended the conference on **Theories, Experiments and Numerics on Gapless Quantum Many-body Systems** held at KITP, Santa Barbara. — May 13-16, 2024

Gave a short talk at the **2024 APS March Meeting, Minneapolis**. A study of dissipative models based on Dirac matrices, Jyotsna Gidugu and Daniel P Arovas. — Mar 4-8, 2024

Gave a talk and presented a poster at the **2023 Annual Meeting of the APS Far West Section held in the University of California, San Diego, California**. A study of dissipative models based on Dirac matrices, Jyotsna Gidugu and Daniel P. Arovas. — Oct 6-7, 2023

Presented a poster at the **Princeton Summer School on Condensed Matter Physics 2023: Fractionalization, criticality and unconventional quantum materials**. A study on dissipative models based on Γ-matrices, Jyotsna Gidugu and Daniel P. Arovas. — Jul 10 - 14, 2023

Attended and presented a poster at the **3rd Condensed Matter Summer School, "Dynamics and Quantum Information in Many-body Systems" at the University of Minnesota**. A study on dissipative models based on Γ-matrices, Jyotsna Gidugu and Daniel P. Arovas. — Jun 12-21, 2023

Presented a poster at a **conference at the Kavli Institute for Theoretical Physics on Topology, Symmetry and Interactions in Crystals: Emerging Concepts and Unifying Themes**. A study on dissipative models based on Γ-matrices, Jyotsna Gidugu and Daniel P. Arovas. — Apr 3-6, 2023

RESEARCH EXPERIENCE

Effects of dissipation in open quantum systems — 2021-Present

Have worked on studying the effects of dissipation on spin and fermionic systems. One of our projects involved studying 2d dissipative models based on Dirac matrices. Currently working on studying the effects of dissipation on the Aubry-André-Harper model.

Under the supervision of **Prof. Daniel P Arovas, Department of Physics, University of California San Diego** as a part of my **PhD research**.

Josephson Junction arrays and high-T_c superconductors 2019 Aug-Present
Using Monte Carlo simulations of a 2D XY model to study magnetic and transport properties of high-T_c superconductors.
Under the supervision of **Prof. H R Krishnamurthy and Prof. Subroto Mukerjee, Centre for Condensed Matter Theory, IISc** as a part of my **Master's dissertation**.

An investigation of thermalization and transport in a generalized Aubry-Andre model 2018 Aug-2019 Jul
Used computational methods such as exact diagonalization to study transport, level spacing statistics and entanglement in a system with a quasiperiodic potential.
Under the supervision of **Prof. Subroto Mukerjee, Centre for Condensed Matter Theory, IISc** as a part of my **Bachelor's dissertation**.

Topological Hall effect in Heusler thin films of Mn_2RhSn 2018 May-2018 Jul
Studied the topological Hall effect in Heusler thin films of Mn_2RhSn to understand conditions that favor skyrmions and antiskyrmions.
Under the supervision of **Prof. Stuart Parkin** during my **summer internship at the Max Planck Institute of Microstructure Physics**, Halle, Germany.

Structural and electronic roperties of $h-BN$ and MoS_2 2017 May-2017 Jul
Studied the structural and electronic properties of some 2D materials like $h-BN$, MoS_2 and their hetero-structures using simulations and micropscopy techniques.
Part of the **Indian Academy of Sciences summer fellowship (Science Academies' Summer Research Fellowship Programme-2017)**.
Under the guidance of **Prof. Ranjan Datta, at International Centre for Materials Science (ICMS), JNCASR**, Bengaluru.

Magnetic and transport properties of some double perovskites with antisite disorder 2016 May-2016 Jul
Studied the structural, magnetic, and magneto-transport properties of $La_{1.5}Sr_{0.5}CoMnO_6$ (LSCMO) thin films.
Under the guidance of **Prof. P S Anil Kumar at the magnetic and spintronic studies lab, IISc, Bengaluru** as a part of my summer project.
Resulted in a publication and a conference paper.

FELLOWSHIPS AND AWARDS

Poster Award Oct, 2023
Won the SPS Poster Award for a graduate student at the 2023 APS Far West Section Annual meeting at UCSD.

Physics Excellence Award at UCSD 2020 Nov-2021 Jul
Awarded in the first-year of duration at UCSD.

Kishore Vaigyanik Protsahan Yojana [KVPY-SA] 2015 Aug-Present
Qualified in the KVPY exam in class 11.
Scholarship program funded by Department of Science and Technology, India.

National Talent Search Examination [NTSE- 2011] 2011 - 2014
Qualified in the NTSE exam in class 8.
Awarded by NCERT, Govt. of India.

RELEVANT COURSES STUDIED

Graduate level:
Classical Mechanics, Quantum Mechanics, Mathematical Physics, Statistical Mechanics, Electrodynamics, Condensed Matter Physics, Nonequilibrium Statistical Mechanics, Field Theory and Renormalization group, Dynamics: Deterministic, Stochastic, Statistical.

Undergraduate level:
Physics courses- Mechanics, Electromagnetism, Quantum Physics, Oscillations and waves, Thermal Physics and Physics of Materials
Math courses- Analysis, Linear Algebra, Probability and Statistics
Engineering courses- Algorithms and Programming, Electrical and Electronics Engineering, Materials Science

TECHNICAL SKILLS

Programming languages:
C, C++, Python

Mathematical packages:
ITensor (used to create tensor network algorithms), MATLAB, Mathematica, WIEN2k (used to study crystal properties)

Computational skills:
Strong knowledge of high-performance computing, MPI Parallel Programming, Joblib, Python multiprocessing.
Data analysis and visualization using Matplotlib, Origin.

Statistics:
Strong knowledge of Monte Carlo simulations, Langevin dynamics, studying statistics of eigenspectra of quantum systems, etc.

OTHER PROGRAMS

Research Science Initiative - 2014
Was among the high school students selected to participate in the Research Science Initiative, Chennai- 2014.
Worked under the guidance of Dr. C V Krishnamurthy in the Department of Physics, IIT Madras during the summer of 2014 on a project on cloud chambers (particle detectors).